Academic Minor Project

Official Technical Documentation

ResQAI: Community-Validated Disaster Intelligence

An Offline-First, Multilingual Geospatial Platform for High-Risk Coastal Geographies

Target Hazard Region: Coastal Belt, River Deltas & Highlands of Odisha, India
Architecture: Offline PWA • Supabase PostgreSQL (RLS) • Deno Edge Functions • Vernacular LLM
Source Repository: [resq-ai-assistant (Local Workspace)] | **Academic Scope:** B.Tech CSE Minor Project

1. Executive Summary & The Elevator Pitch

> ***The 2-Minute Pitch for Sir:***
> *"ResQAI is an offline-capable, community-validated disaster intelligence and emergency response platform built for vulnerable coastal geographies like Odisha, India. Unlike conventional centralized alert platforms that collapse during network blackouts or broadcast unverified rumors, ResQAI introduces a bi-directional feedback loop: satellite, seismic, and weather feeds trigger predictive risk models, while on-ground citizens validate the hazard with a single tap, dynamically updating an automated consensus trust score. Powered by an installable offline-first Progressive Web App (PWA), ResQAI ensures life-saving advisories, turn-by-turn shelter routing, and voice-assisted AI emergency guidance reach the last mile in Odia, Hindi, and English."*

```
[Multi-Sensor Feeds] ⟶ AI Predicts Risk ⟶ Citizens Tap Confirm/Deny ⟶ Trust Score Dynamically Calibrates ⟶ Tiered Emergency Alerts
```

2. Problem Statement & Regional Context (Odisha Focus)

Coastal Odisha is one of the most natural-disaster-prone regions globally. Flanked by the warm waters of the Bay of Bengal, the state faces recurrent extreme tropical cyclones (1999 Super Cyclone, Phailin 2013, Fani 2019, Dana 2024), riverine flooding across the Mahanadi and Brahmani delta plains, heatwaves in western districts (Sambalpur, Balangir), and landslides in the Eastern Ghats.

Legacy disaster management systems exhibit 5 structural failures during peak crises:

- **The Last-Mile Network Blackout:** High-velocity winds and water inundation snap fiber lines and drain cell tower battery banks, rendering online portals unreachable.
- **Rumors and Information Asymmetry:** Official automated sensors are spaced hundreds of kilometers apart; social media spreads unverified rumors, while genuine hyper-local flash floods go unnoticed.
- **The "Black-Box" AI Dilemma:** Responders and district magistrates distrust opaque percentage scores that fail to explain *why* an area is flagged as high-risk.
- **Language & Literacy Barriers:** Bulletins are predominantly in English or formal administrative Hindi, excluding rural coastal citizens whose mother tongue is Odia.
- **Opaque Shelter Capacities:** Evacuees have no real-time visibility into whether a designated cyclone shelter is already overflowing or safe to access.

3. System Architecture & Technical Specifications

ResQAI implements a decoupled, offline-resilient architecture spanning six distinct operational layers:

Architectural Tier	Technologies Deployed	Direct Link	Role & Architectural Responsibility
Client Interface (PWA)	React 18, TypeScript, Vite 7, Tailwind CSS, shadcn/ui	react.dev vite.dev	Responsive emergency command interface, glassmorphism telemetry panels, and sub-second rendering.
Geospatial Engine	Leaflet 1.9, CARTO Dark Basemaps, OSRM Routing Machine	leafletjs.com project-osrm.org	Renders 15-station monitoring grid across Odisha, canvas risk heatmaps, and turn-by-turn walking/driving routes.
Offline & Caching	vite-plugin-pwa (Workbox), Service Workers, localStorage	vite-pwa-org	Precaches static application shell, map tile caching, and offline fallback heuristic risk calculations when disconnected.
Backend as a Service	Supabase PostgreSQL, Row-Level Security, Realtime WebSockets	supabase.com/docs	Stores citizen reports, shelter statuses, and risk predictions. Enforces RBAC (Citizen, Responder, Admin).
Serverless Edge Layer	Deno Runtime (Edge Functions: predict-risk, resqai-chat)	deno.land	Isolates AI inference and telemetry aggregation. Validates JWT access tokens before consuming AI token quotas.
Live Telemetry Feeds	NASA EONET, USGS Earthquakes, Open-Meteo, NASA FIRMS	eonet.gsfc.nasa.gov earthquake.usgs.gov	Real-time ingestion of seismic events, active wildfire thermal hotspots, atmospheric pressure, precipitation, and wind speeds.

The 4-Tier Early Warning Alert Model

Emergency alerts in ResQAI follow a strict graduated hierarchy to avoid crying-wolf syndrome while ensuring critical events trigger immediate audio and visual warnings:

Alert Tier	Trust / Confidence	Trigger Condition	Action Enforced
`VERIFIED_CRITICAL`	Confidence ≥ 0.85 Trust ≥ 0.75	AI high prediction + community ground consensus or official admin verification	Pulsing red radar ring on map, auto-cycling banner, mandatory shelter evacuation route, local audio warning tones.
`AI_PREDICTED`	Confidence ≥ 0.70	Multi-factor meteorological threshold breached (e.g. 7d rain > 180mm, wind > 85 km/h)	Orange alert banner, pushes verification prompt to citizens in affected grid cell to confirm ground truth.
`MONITORING`	Confidence 0.40 - 0.69	Sub-critical atmospheric pressure drop or disputed community reports	Yellow telemetry badge on map station, live sensor poll frequency elevated from 30m to 5m.
`LOW_WATCH`	Confidence < 0.40	Standard seasonal baseline conditions	Green status pin, normal telemetry logging.

4. Core Functional Modules (Part I: Mapping, Prediction & Trust)

Module 1: Geospatial Disaster Command Map

The central console is built with Leaflet over dark CARTO basemaps, optimized for high visibility during nighttime power cuts. The system continuously evaluates a 15-station monitoring grid covering high-risk hubs across Odisha: Bhubaneswar, Cuttack, Puri Coast, Paradip, Balasore, Kendrapara, Jajpur, Mayurbhanj, Berhampur, Koraput, Sambalpur, Rourkela, Balangir, Boudh, and Dhenkanal.

- **Canvas Risk Heatmaps:** 6 selectable layers visualizing flood inundation, cyclone track intensity, active wildfire, seismic tremor, landslide susceptibility, and heatwave anomalies.
- **Animated Radar Pulse Rings:** Grid points flagged with `CRITICAL` or `HIGH` alert status emit pulsing concentric rings.
- **Live Shelter Markers:** Color-coded according to occupancy capacity (Green < 70%, Yellow 70-90%, Red > 90%).

Module 2: Risk Prediction & Explainable AI (XAI)

ResQAI implements a dual-mode prediction pipeline. If an external model endpoint (`ML_MODEL_URL`) is configured, the system pipes multi-parameter vectors into that model. Otherwise, the edge function executes a calibrated heuristic risk engine.

Explainability Breakdown: Rather than providing an opaque probability, every prediction exposes quantified environmental factors:

- *Meteorological Factors:* 7-day cumulative rainfall (> 180 mm), max sustained wind velocity (> 85 km/h), barometric pressure drop.
- *Geographic Constants:* Distance to Bay of Bengal coastline (< 25 km), low-lying Mahanadi delta coordinates, Simlipal forest reserve density, steep slope gradients in Koraput highlands.
- *Seasonal Multipliers:* Indian monsoon window (June–September) and peak post-monsoon cyclone season (October–November).

Module 3: Community Validation & Algorithmic Trust Scoring

To prevent false alarms and rumors, ResQAI pairs AI risk predictions with on-ground crowd validation:

- Citizens within an affected grid point can tap **"Confirm"** or **"Deny"** on active alerts or peer incident reports.
- The system dynamically computes the consensus trust score:

```
Trust Score = Confirmations / (Confirmations + Denials)
```

- Alerts with high confidence (> 0.75) escalate to `VERIFIED_CRITICAL`. Reports with low trust (< 0.40) are downgraded to `MONITORING` or suppressed.

4. Core Functional Modules (Part II: AI Assistant, Routing & Offline Resilience)

Module 4: Vernacular Emergency AI Assistant (`resqai-chat`)

Provides immediate life-safety advice (cyclone evacuation procedures, flood safety, snakebite first aid, medical triage) with strict language enforcement:

- **Strict Vernacular Guardrails:** System prompts mandate responses exclusively in the chosen language script: Odia script (ଓଡ଼ିଆ), Devanagari Hindi, or English. English mixed into Odia responses is strictly rejected at the prompt boundary.
- **Voice Input Support:** Integrates the browser's Web Speech API for hands-free speech-to-text queries, supporting users in acute distress or with lower literacy.
- **Authentication Gate:** The Edge Function inspects the user's Supabase JWT token before contacting LLM providers, preventing credit depletion by unauthenticated bots.

Module 5: Evacuation Routing & Shelter Logistics

Evacuation navigation is powered by Leaflet Routing Machine integrated with the Open Source Routing Machine (OSRM):

- **Automated Nearest Safe Shelter Matching:** Computes distance from user GPS coordinates to all registered shelters in the database.
- **Occupancy Awareness:** Excludes shelters where occupancy has reached 100%, re-routing evacuees to the next nearest safe facility.
- **Multi-Modal Profiles:** Calculates distinct routes for walking (footpath paths avoiding major river surges) and driving (high-ground road corridors).

Module 6: Offline Resilience & PWA Architecture

When mobile cellular towers fail, ResQAI ensures uninterrupted local execution:

- **Service Worker (Workbox):** Precaches the complete application shell, JavaScript runtime, CSS design tokens, icons, and vector basemap assets.
- **Runtime Tile Caching:** Map tiles traversed by the user are cached in the browser's CacheStorage API for offline rendering.
- **Client Heuristic Fallback:** If network requests fail, ResQAI pulls the latest cached grid from `localStorage` and computes risks locally on-device.
- **Local Audio Synthesis:** Emergency warning sirens are generated on-device via the Web Audio API without requiring remote audio file streaming.

5. Complete Database Schema & Data Models (Supabase PostgreSQL)

The backend leverages PostgreSQL on Supabase with strict Row-Level Security (RLS) policies protecting table integrity:

Table Name	Key Columns & Data Types	Primary & Foreign Keys	Row-Level Security (RLS) Policy
`profiles`	id (uuid), full_name (text), language (varchar), phone (text), emergency_contact (text)	PK: id FK: auth.users.id	Users can SELECT & UPDATE only their own profile row (`auth.uid() = id`).
`user_roles`	id (uuid), user_id (uuid), role (text: 'citizen' | 'responder' | 'admin')	PK: id FK: auth.users.id	Readable by authenticated users; only system admins can insert or modify role assignments.
`reports`	id (uuid), user_id (uuid), disaster_type (text), description (text), lat (float), lng (float), trust_score (float), verified (bool)	PK: id FK: profiles.id	All users can SELECT active reports; authenticated users can INSERT; only responders/ admins can set verified=true.
`report_validations`	id (uuid), report_id (uuid), user_id (uuid), is_valid (boolean), created_at (timestamptz)	PK: id FK: reports.id	Citizens can submit one vote per report; duplicate votes by the same user on the same report are rejected.
`risk_predictions`	id (uuid), grid_id (text), disaster_type (text), risk_level (text), confidence (float), factors (jsonb)	PK: id	Publicly readable by all client sessions; insert/ update restricted to Edge Function service role.
`shelters`	id (uuid), name (text), lat (float), lng (float), capacity (int), occupancy (int), amenities (jsonb)	PK: id	Publicly readable for evacuation navigation; only responders/admins can update occupancy figures.
`alerts`	id (uuid), title (text), message (text), severity ('CRITICAL'|'HIGH'|'MED'|'LOW'), region (text)	PK: id	Publicly readable; real-time changes broadcast over Supabase WebSockets to all connected clients.
`resources`	id (uuid), item_name (text), category (text), quantity (int), location (text)	PK: id	Restricted to responder and admin roles for disaster logistics (boats, medical supplies, food packets).

Security & Token Architecture

ResQAI implements dual-layer protection: (1) **API Gateway Protection:** The public anon key is read-only for public tables and insert-only for personal reports. (2) **JWT Bearer Validation:** Deno edge functions verify the signature of the caller's JWT before executing AI queries, completely blocking anonymous scrapers.

6. Peer-Reviewed Research Papers & Literature Survey

The table below provides direct active digital object identifier (DOI) links to peer-reviewed academic literature validating ResQAI:

Author(s) & Year	Publication & Publisher	Paper Title & Direct DOI Link	Core Relevance to ResQAI Architecture
Michael F. Goodchild (2007)	GeoJournal *Springer*	**Citizens as sensors: the world of volunteered geography** **https://doi.org/10.1007/s10708-007-9111-y**	Seminal paper establishing Volunteered Geographic Information (VGI). Directly justifies ResQAI's citizen validation model where humans act as distributed spatial sensors.
Muhammad Imran et al. (2015)	ACM Computing Surveys (CSUR) *ACM / IEEE*	**Processing Social Media Messages in Mass Emergency: A Survey** **https://doi.org/10.1145/2771588**	Identifies the noise and rumor challenge in crisis crowdsourcing. Backs ResQAI's structured confirm/deny voting mechanism and trust calculation.
Saman Ghaffarian et al. (2023)	Int. Journal of Disaster Risk Reduction *Elsevier*	**Explainable artificial intelligence in disaster risk management** **https://doi.org/10.1016/j.ijdrr.2023.104123**	Demonstrates that emergency incident commanders reject "black-box" models. Directly justifies ResQAI's factor-level explainability breakdown.
Toushif Jaman et al. (2021)	Int. Journal of Disaster Risk Reduction *Elsevier*	**Assessment of impact of cyclone hazard on social vulnerability in Odisha** **https://doi.org/10.1016/j.ijdrr.2020.101997**	Empirical case study of coastal Odisha during Cyclones Phailin and Titli. Emphasizes the need for localized warnings and real-time shelter routing.
IEEE Conference (2021)	IEEE IoTaIS Proceedings *IEEE Xplore*	**Disaster Management Application for After-Disaster Rapid Mobile Assessment** **https://doi.org/10.1109/IoTaIS50849.2021.9359695**	Validates the requirement for offline-first mobile web architectures that continue operating under severed telecommunications.

Synthesis of Academic Findings

These papers confirm that successful disaster tech requires combining **machine prediction with human-in-the-loop validation** (Goodchild 2007, Imran 2015), **transparent explainability** (Ghaffarian 2023), and **offline-resilient communication** (IEEE 2021) applied to high-risk coastal topographies like Odisha (Jaman 2021).

7. Top Viva Voce Examination Questions & Model Answers

Q1: Why did you choose coastal Odisha as your focus area rather than a generic national portal?

Answer: "Sir, coastal Odisha is India's most cyclone-vulnerable state, with historical events like the 1999 Super Cyclone, Phailin, and Fani. While macro-level IMD bulletins exist, local disaster mortality occurs because of the 'last-mile' gap: telecommunication blackouts, language barriers, and lack of real-time shelter vacancy tracking. Building for Odisha allowed us to tailor geographical constants (Mahanadi delta, coastal coordinates) and regional language models (Odia)."

Q2: How does the app operate when cellular networks fail completely during a cyclone?

Answer: "Sir, ResQAI employs an offline-first PWA architecture via Workbox Service Workers. It precaches the application shell, scripts, CSS, and map tiles. When the device loses connection, the app switches to cached predictions stored in localStorage and evaluates local fallback heuristics. Alerts trigger locally synthesized Web Audio tones without requiring media streaming."

Q3: How do you prevent false rumors or panic from crowdsourced reports?

Answer: "Sir, we employ a 3-layer security model: (1) All reports require authenticated sessions; anonymous submissions are rejected. (2) Every report starts in an unverified state; nearby citizens submit confirm/deny votes, calculating a consensus trust score: Confirmations / (Confirmations + Denials). Reports with low consensus are automatically suppressed. (3) Official emergency responders can review and flag reports in the Admin Panel."

Q4: What makes your risk prediction 'Explainable' (Explainable AI)?

Answer: "Sir, emergency responders distrust black-box algorithms. Instead of showing an opaque risk percentage, ResQAI displays explicit contributing environmental factors such as 7-day cumulative precipitation > 180mm, sustained wind gusts > 85 km/h, and coastal delta proximity."

Q5: What is Row-Level Security (RLS) in your database?

Answer: "Sir, Row-Level Security in PostgreSQL ensures that database queries are restricted at the database engine level based on the caller's JWT role. Regular citizens can only read active alerts and insert their own incident reports. Only accounts with the 'responder' or 'admin' role in user_roles can update shelter capacities or clear incident queues. This prevents API tampering even if the frontend anon key is inspected."

Q6: How do you enforce strict Odia and Hindi output in the AI Chatbot?

Answer: "Sir, our 'resqai-chat' Edge Function injects strict system prompts requiring the language model to generate output strictly in the requested vernacular script (Odia unicode block or Devanagari). It also gates execution behind an auth check, preventing anonymous token drain."

Q7: What routing mechanism is used for cyclone shelter evacuation?

Answer: "Sir, we integrate Leaflet Routing Machine with OSRM (Open Source Routing Machine) to compute real-time walking and driving paths to the nearest shelter that has available capacity."

8. Five-Minute Live Demonstration Script for Sir

Time Window	Module & Screen Route	Actions to Perform on Screen	What to Say to Sir (Verbal Cue)
0:00 - 0:45	Landing Screen (`/`)	Open homepage, show problem statement and architecture badges.	*"Sir, this is ResQAI. It is built to address the last-mile communication and validation breakdown during coastal cyclones in Odisha."*
0:45 - 1:45	Command Map (`/dashboard`)	Pan the 15-station Odisha grid. Toggle between Cyclone, Flood, and Fire heatmaps.	*"Here is our Leaflet Command Console. Notice the 15 sensor stations and the pulsing rings around active high-risk zones like Puri Coast and Paradip."*
1:45 - 2:30	Forecast Center (`/forecast`)	Hover over a high-risk marker to reveal the Explainability tooltip.	*"Rather than giving an opaque probability, our Explainable AI breaks down the contributing factors: 7-day rainfall, wind velocity, and coastal proximity."*
2:30 - 3:15	Community Validation	Cast a Confirm vote on an active community report.	*"Here citizens act as distributed sensors. As confirmations increase, the Trust Score recalculates live via Supabase WebSockets, filtering out fake rumors."*
3:15 - 4:00	Vernacular AI Assistant	Toggle language to Odia / Hindi. Query evacuation steps with voice input.	*"Our emergency assistant generates strict Odia and Hindi instructions, with voice-input support for citizens in crisis."*
4:00 - 5:00	Shelter Routing (`/evacuate`) & Admin	Calculate route to nearest cyclone shelter; show admin triage.	*"Finally, the platform navigates evacuees to the nearest active shelter with capacity, while giving administrators an incident triage control room."*

9. Limitations & Roadmap for Final Year (Major Project)

- **IoT LoRaWAN River Sensor Grids:** Deploying physical ultrasonic river-depth nodes along the Mahanadi delta that transmit telemetry over long-range, battery-powered radio frequencies when cellular towers collapse.
- **Satellite Synthetic Aperture Radar (SAR) Inundation Mapping:** Processing automated flood extent vectors using European Space Agency (ESA) Sentinel-1 radar imagery via Google Earth Engine.
- **Device-to-Device Bluetooth Low Energy (BLE) Mesh Relay:** Enabling smartphones in an affected village to relay SOS messages and emergency alert packets phone-to-phone without cell towers.
- **On-Device Quantized SLM:** Packaging a small language model (e.g. Gemma 2B or Llama-3-8B) to run entirely within the browser via WebGPU / ONNX Runtime for completely offline conversational triage.